Enrollment No.: _____

Darshan UNIVERSITY

योगः कर्मसु कौशलम्

Darshan Institute of Computer Application

BCA / B.Sc. (IT) | Sem-5 | Winter-2023

Course Code	:	2104CS521	**Date**	:	03-11-2023
Course Name	:	Introduction to Advanced Technologies	**Duration**	:	150 Minutes
			Total Marks	:	70

Instructions:

1. Attempt all the questions.
2. Figures to the right indicates maximum marks.
3. Make suitable assumptions wherever necessary.

Q.1	**(A)**	Define Cloud Computing. List operations of cloud computing.	**4**
	(B)	Compare cloud computing and grid computing.	**3**
		OR	
		Compare public cloud and private cloud.	
	(C)	Explain cloud computing architecture with diagram.	**7**
		OR	
		Explain cloud service model (IAAS, PAAS, SAAS).	
Q.2	**(A)**	What is virtual reality? Explain virtual reality in details.	**4**
	(B)	Write the benefits of virtual reality and augmented reality.	**3**
		OR	
		Write the disadvantages of virtual reality and augmented reality.	
	(C)	Discuss the challenges in augmented reality and virtual reality.	**7**
		OR	
		What is augmented reality? Explain augmented reality in details.	
Q.3	**(A)**	Define Artificial Intelligence and Machine Learning with example.	**4**
	(B)	Write a short note on supervised learning in M/L.	**3**
		OR	
		Write a short note on unsupervised learning in M/L.	
	(C)	Explain different phases of Machine Learning Model workflow.	**7**
		OR	
		Discuss different rea-life applications of Machine Learning.	

Q.4	**(A)**	Explain the data science pipeline.	**4**
	(B)	Explain python's role in data science.	**3**
		OR	
		What is dataframe? Explain structure of dataframe with an example.	
	(C)	Explain python ecosystem for Data Science.	**7**
		OR	
		Explain xs() function with an example.	
Q.5	**(A)**	Define blockchain technology with its applications.	**4**
	(B)	Write a short note on Proof of Work (PoW).	**3**
		OR	
		Write a short note on Proof of Stack (PoS).	
	(C)	Explain blockchain architecture with proper diagram.	**7**
		OR	
		Explain any two types of blockchain with example.	

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